

HAT-P-36b

R-0970

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_230123 · created 2026-07-29T17:26:55+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459967.999 +/- 0.005 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.156 +/- 0.044
Transit depth (Rp/Rs)^2	2.43 +/- 1.37 [%]
Orbital Inclination (inc)	83.32 +/- 2.81
Airmass coefficient 1 (a1)	28.81 +/- 14.67
Airmass coefficient 2 (a2)	-2.98 +/- 1.17
Scatter in the residuals of the lightcurve fit is	52.48 %
Best Comparison Star	#3 - [530, 157]
Optimal Aperture	4.12
Optimal Annulus	5.71
Transit Duration (day)	0.088 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.156 ± 0.044 vs 0.126 (deviation 0.59σ)
Duration fit vs published	0.088 d vs 0.0925 d (deviation 0.32σ)
Mid-time O–C	11.1 min
Depth SNR	3.0
Residual scatter	52.48 %

Light curve

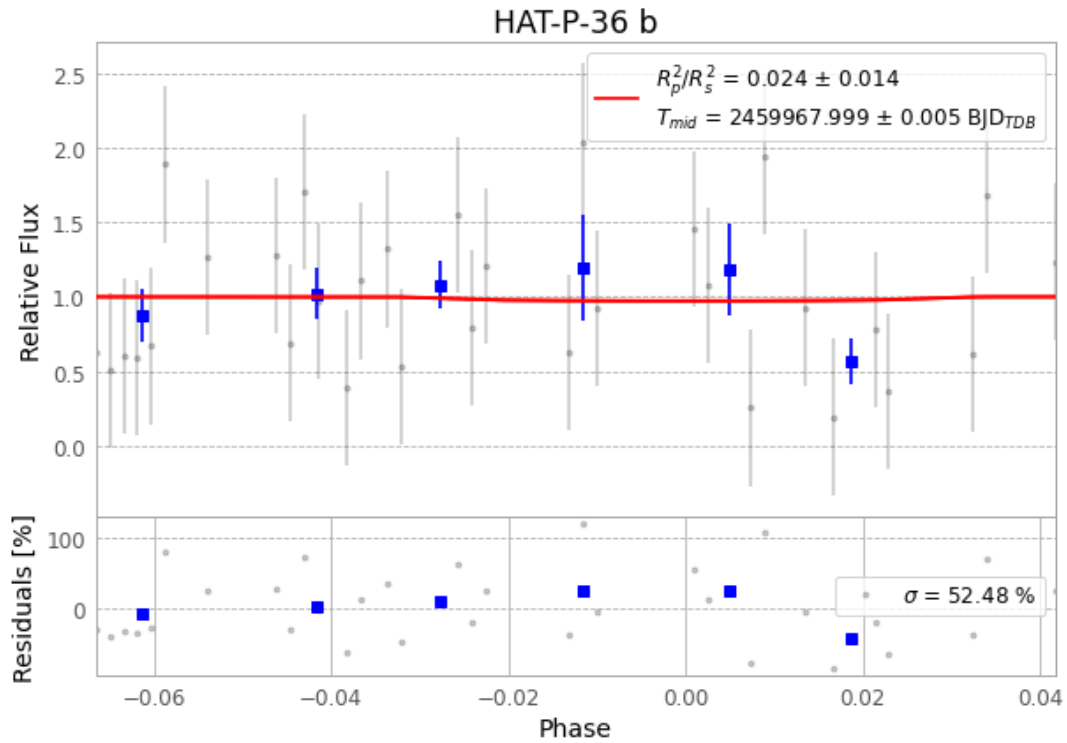


Figure 1 — FinalLightCurve_HAT-P-36 b_2023-01-23.png (SHA-256 715dff14e15cdeca...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [537, 173], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 47a03070f0c8a326...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

75 FITS frames (49 MB), HATP-36230123094516.FITS → HATP-36230123132715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-230123.log (d5bcf5f00169...), FinalLightCurve_HAT-P-36 b_2023-01-23.png (715dff14e15c...), FinalLightCurve_HAT-P-36 b_2023-01-23.csv (ccce9f06b5c9...)

Compute resources

Wall time 225.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 17:26Z.